Guide 1. APT Project Definition Capstone Course

A. PART I

1. Personal Information

Below is a table where you must complete the requested information.

Student Name	**Osvaldo Antonio Reyes Wiegand - Lucas Joaquin Carvajal Pincheira**
National ID (RUT)	**19.336.327-k / 21.146.055-5**
Degree Program	**Computer Science & Software Engineering (Ingeniería en Informática)**
Campus	**Viña del Mar**

2. APT Project Description

In this description, briefly state the name of your APT project and the graduate profile competencies you will put into practice. If performance areas are defined in your degree program, also mention which performance areas the project is linked to.

Project Name	*Predictive Demand and Inventory Optimization System using Machine Learning for the Wholesale Textile and Footwear Sector.*
Performance Area(s)	*Software Development, Database Management, and Artificial Intelligence.*
Competencies	*Design and generate innovative, high-quality software solutions by applying the software lifecycle.* *Design and generate solutions that resolve organizational information requirements, considering relational databases.* *Design and adapt requirements engineering processes for the development of complex IT solutions.*

3. APT Project Rationale

Below are different fields that you must complete with the requested information. This section aims to describe your project in detail and justify its relevance and pertinence.

Relevance of the APT Project	The wholesale and retail textile and footwear sector constantly faces the problem of overstocking (capital tied up in "dead stock") or stockouts during high-demand seasons. This project aims to solve this inefficiency through accurate predictions. The problem is located in the Valparaíso Region, specifically impacting the logistics and finances of a local wholesale company in Quilpué. Its resolution provides tangible business value by transforming empirical decision-making into a data-driven strategy.
APT Project Description	The goal is to optimize the client's seasonal purchasing. The project consists of developing a Data Warehouse to centralize historical sales data and building a Machine Learning model (time series and regression algorithms) capable of predicting future demand volumes for apparel and footwear, considering temporal and business variables.
Alignment with the Graduate Profile	This project requires fully applying the graduate profile competencies by demanding formal requirements gathering with a real client, modeling and building a structured relational/dimensional database, and programming an innovative software solution (incorporating Artificial Intelligence) under strict quality standards.
Alignment with Professional Interests	This project reflects our professional interest in backend development, data architecture, and artificial intelligence algorithm programming. It contributes to professional growth by integrating tools such as Python and robust database engines to solve a critical, real-world business problem.
Feasibility of the APT Project	The project is feasible to develop within the semester duration and allocated hours. Required material resources are accessible (development environments such as Visual Studio Code, Python, and database engines such as SQL Server for managing the dimensional model). A facilitating factor is having the company's tax ID (RUT) and active cooperation. The main factor that could hinder development is the initial dispersion of the client's historical records; this is addressed by implementing an automated ETL pipeline to structure and clean the client's sales dataset in order to effectively train the Machine Learning model.

B. PART II

4. Objectives

In this section, define the general and specific objectives of the APT Project. Objectives must be stated clearly, concisely, and without further explanation so that they are self-explanatory. Writing them using infinitive verbs is recommended to specify concrete actions.

General Objective	To develop a predictive software system based on Machine Learning to optimize inventory management and estimate product demand in a textile and footwear wholesale company.
Specific Objectives	To gather and model the functional and non-functional requirements of the predictive system alongside the client. To design and implement a dimensional model (Data Warehouse) for historical consolidation of inventory and sales data. To train and evaluate a Machine Learning model using Python for product demand forecasting. To build a user interface that allows the client to visualize predictions and manage inventory alerts.

5. Methodology

In the following section, describe the discipline-specific methodology you will use to carry out the APT project described above, including stages and working methods.

Methodology Description
To address the stated problem and ensure the fulfillment of objectives, the project will be developed under the Agile Scrum framework. This choice is justified by the iterative and incremental nature of Artificial Intelligence software development, allowing continuous product adaptation and model refinement as the project progresses. ***Osvaldo Reyes (Technical Lead & Developer): Directly responsible for the technological construction of the solution. Main tasks include designing and implementing the dimensional model (Data Warehouse) using SQL Server, programming and training Machine Learning algorithms in Python, and developing the business logic and software interface.*** ***Lucas Carvajal (Documentation & QA Lead): Responsible for requirements management, documentation quality, and project tracking. Main tasks include gathering and drafting requirements (User Stories), creating software engineering and architecture diagrams, writing technical progress and final reports, and supporting Product Backlog management to ensure alignment with academic standards.***

6. Deliverables and Evidence

Below, describe the evidence that will be evaluated in the progress report and final report of your APT project. These deliverables must be agreed upon with your instructor. Evidence refers to the artifacts developed during the project intended to demonstrate and document how the work was implemented.

Evidence Type (Progress or Final)	Evidence Name	Description	Justification
Progress	**Data Cleaning Pipeline and Dimensional Model (ETL)**	*Python script and database design (SQL Server) that extracts, processes, and structures the historical sales and catalog of the textile company.*	**Justified through a Data Quality KPI. It proves that 100% of null or outlier values were treated, ensuring an integral and structured database so that the predictive algorithm can be trained without errors.**
Progress	**Trained Predictive Algorithm (Machine Learning)**	**Source code developed in Python demonstrating data cleaning, predictive algorithm training, and accuracy metrics.**	**Justified through a Model Precision KPI. It mathematically proves that the AI predictions have an acceptable margin of error useful for business decision-making.**
Final	**Deployed Functional Predictive Software**	**Complete software system integrating the database, AI engine, and user interface to visualize demand predictions.**	**Justified through a Performance and Acceptance KPI. It demonstrates that the complete system operates under engineering standards, delivering predictions to the user within optimal response times (low latency) and fulfilling all final Sprint user stories.**
			KPI-BASED JUSTIFICATIONS

7. Work Plan

In the following table, define the planning of your APT Project as required.

APT Project Work Plan						
Competency or Competency Units	Activity / Task Name	Activity / Task Description	Resources	Activity Duration	Responsible[1]	Observations / Remarks
Design requirements engineering processes.	*Requirements gathering.*	*Meeting with the client to define inventory variables.*	*Virtual / in-person meeting.*	*1 week*	*Lucas Carvajal*	*Coordinate schedule with the client.*
Design solutions with relational databases.	*Data Warehouse design.*	*Creation of fact and dimension tables in SQL Server.*	*SQL Server Engine, PC.*	*2 weeks.*	*Osvaldo Reyes*	*Depends on dataset completeness.*

[1] En caso de que el Proyecto APT sea grupal, en esta columna deben indicar el nombre de los responsables de cada tarea o actividad. Esto posteriormente permitirá diferenciar la evaluación por cada integrante.

8. Gantt Chart

Select a suitable Gantt Chart format and organize the planned activities considering the period assigned for your APT Project development. Maintain the academic timeline across the three phases of the Capstone Portfolio Course.

Activity	Phase 1				Phase 2											Phase 3		
	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16	W 17	W 18
Initial meeting and requirements gathering (Milestone 1) (Mitigates: lack of formal requirements)	X																	
Architecture definition and data strategy (Mitigates: unstructured historical records)		X																
Dimensional model design (facts and dimensions) (Mitigates: transactional data dispersion)			X															
Creation and implementation of the Data Warehouse in SQL Server (Mitigates: lack of analytical repository)				X														
ETL pipeline, data cleaning, and validation (Milestone 2 / Progress 1) (Mitigates: inconsistencies, nulls, and outliers)					X													
Python setup and Data Warehouse connectivity						X												

(Mitigates: disconnection between data and AI layers)																	
Feature engineering for seasonality, weather, and holidays (Mitigates: unmeasured seasonality impact)							X										
Training of predictive models (Random Forest / Time Series) (Mitigates: empirical and intuitive purchasing)								X	X								
MAE/MAPE/RMSE evaluation and optimization (Milestone 3 / Progress 2) (Mitigates: high forecasting error margin)										X							
API development and replenishment logic (Mitigates: capital tied up in dead stock)											x	x					
Design and layout of the web interface / Dashboard (Mitigates: difficulty in interpreting AI outputs)												X	X				
Stockout prevention alert module (Mitigates: stockouts during key seasons)													X	X			
Full end-to-end technical integration (SQL -> AI -> API -> UI) (Mitigates: component fragmentation)															X		
Stress testing and operational validation with the client (Mitigates: model misalignment against real demand)																X	
Preparation of technical/user manuals and KPI calculation (Mitigates: lack of methodological backing)																	X
Final system demonstration and project closure																	X

(Milestone 4) (Mitigates: resistance to use or lack of adoption)																		